Jitter/wander analysis of adaptive clock recovery for constant bit rate services over ATM

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One of the problems associated with ATM is the variable transport delay introduced by the cell switches and through waiting time jitter which can perturb any constant bit rate stream traversing the network. An associated paper [1] introduces the three techniques that are available to counter cell delay variation (CDV) and reconstitute constancy of the bit stream exiting the network. This paper concentrates on the effects of varying network load on the adaptive clock recovery technique. The inability of adaptive clock recovery to filter out very low frequency delay changes in an ATM network leads to the recommendation that adaptive clock recovery can only be used for wander-insensitive CBR services.

1. Introduction

Following on from simulation results of 2 Mbit/s adaptive clock recovery [1] where network load characteristics had been found to directly influence the 2-Mbit/s output jitter and wander characteristics, measurements were performed on an ATM switch and 2-Mbit/s adaptive clock recovery module at BT Laboratories. The results were found to match the characteristics of the simulation model and cast doubt on the ability of an ATM network using single priority queue ATM switches to meet the output wander characteristics defined by the ITU Recommendation G.823 specification. The considerations here do not address additional concerns of waiting time jitter which should form the basis of future work.

2. Initial simulation results

The numerical simulation model (see Mulvey, Kim and Reid [1]) had shown the 2-Mbit/s output clock to track the slow changes in switch delay due to changing network load. The model had shown that for an increase in output port load from 70% to 90% the rms switch delay, for an STM-1 output port, would increase by 11.9 ms and that the 2 Mbit/s output clock phase would track this increased rms delay as shown in Figs 1 and 2 respectively. (It should be noted that 11.9 µs = 24.4 unit intervals (UI = bit periods) at 2.048 Mbit/s.)

The mean (rather than rms) switch delay at 90% load exhibited by the numerical model in Fig 1 was 10.1ms. This result is supported by analytical analysis for an M/D/1 queue [2] which gives the worst-case mean delay, $\lambda(\rho)$ in switch cell times, for a given load (r) as:

$$\left(\lambda(\rho) = \frac{\rho}{2(1-\rho)}\right) \qquad \text{...... (1)}$$

Thus, by equation (1), the increase in mean delay expected from a load of 90% = $\lambda(0.9)$ = 4.5 cells = 4.5*2.83 ms for an STM-1 output port = 12.75 µs = 26 UI at 2.048 MHz.

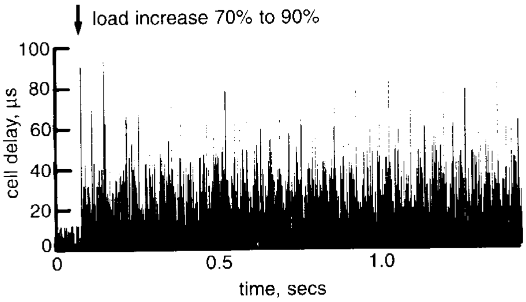


Fig 1 Increase in cell delay.

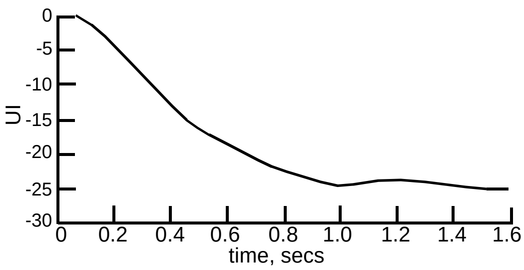


Fig 2 Phase tracking of the adaptive clock recovery mechanism.

It was expected therefore that, if an output port of the ATM switch could be artificially loaded to, say, 90%, phase tracking similar to that shown in Fig 2 ought to occur on the 2-Mbit/s CBR output of the adaptive clock recovery module, thus causing the 2-Mbit/s output phase to wander in concert with longer term (sub-1 Hz) load variations.

3. Experimental configuration

In order to artificially load an output port onto the ATM switch the technique of 'tromboning' was used, whereby the input load source is recirculated through the switch with the relevant VP identity changes and physical patch cord links to cause 'real' load-like contention at a designated output port. Although tromboning is thought to provide imprecise cell delay distributions it was expected that, for a given trombone induced load, roughly the same mean delay increase would be experienced as for 'real' bursty traffic. The tromboning as implemented is shown in Fig 3.

The logical VP connectivity to provide load contention is shown in Fig 4.

As shown in Fig 4 the background load comprised two components. The first was a bursty Poisson load generated by the HP broadband tester to give a mean load of 18.1% on its STM-1 output interface. This load is tromboned through five separate input ports, ports 1 to 5, to give a total load contention at output port No. 1 of 5*18.1 = 90.5%. The second component of the background load was a 2-Mbit/s CBR test stream also generated by the HP broadband tester concurrently with the Poisson load. This test stream was used to measure the mean delay change through the switch as the Poisson load component was turned on and off, for comparison with that predicted by equation (1) and the measured phase change of the 2-Mbit/s receive clock.

The two 2-Mbit/s cell streams from the broadband tester and 2-Mbit/s transmission interface constitute an additional load of $\sim(2*2*10^6/155.52*10^6)*(2430/2340)$ = 2.7%, accounting for STM-1 section and path overhead. Therefore, as the Poisson background load is turned on, the total switch load becomes 90.5 + 2.7 = 93.2%

The worst-case expected delay increase $\lambda(\rho)$ for ρ = 0.932 given by equation (1):

= 6.85 STM-1 cell times

= 6.85 *2.8 μs = 19.2 μs

= $19.2*10^{-6}*2.048*10^6$ = 39.3 unit intervals.

It was therefore expected that the received 2-Mbit/s bit stream from the transmission interface would exhibit up to a 39 UI phase shift on load turn-on. Figure 5 shows the test set-up used to measure this.

The traces shown on the oscilloscopes in the block schematic of Fig 5, and further depicted in Figs 6 and 7, discussed later in section 4.2, show the characteristics observed on load switching.

The 'received data' trace on the dual beam oscilloscope (Fig 5) was seen to track both right and left relative to the 'transmit data' trace as the background Poisson load was turned on and off, highlighting the induced phase change in the receive clock. The question is by how many clock cycles does the receive clock move relative to the transmit clock (i.e. what is the induced peak-to-peak phase wander due to network load change).

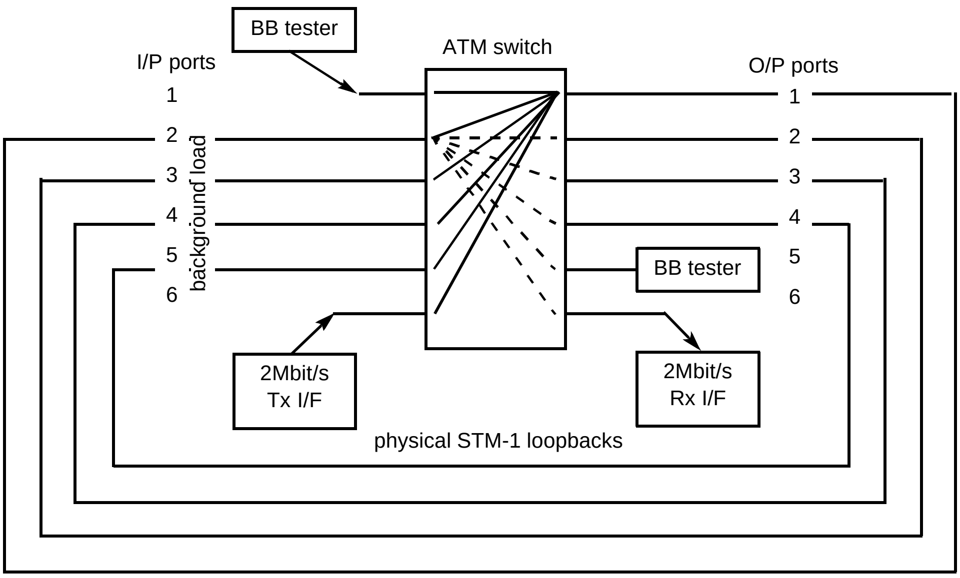


Fig 3 Tromboning of switch connections.

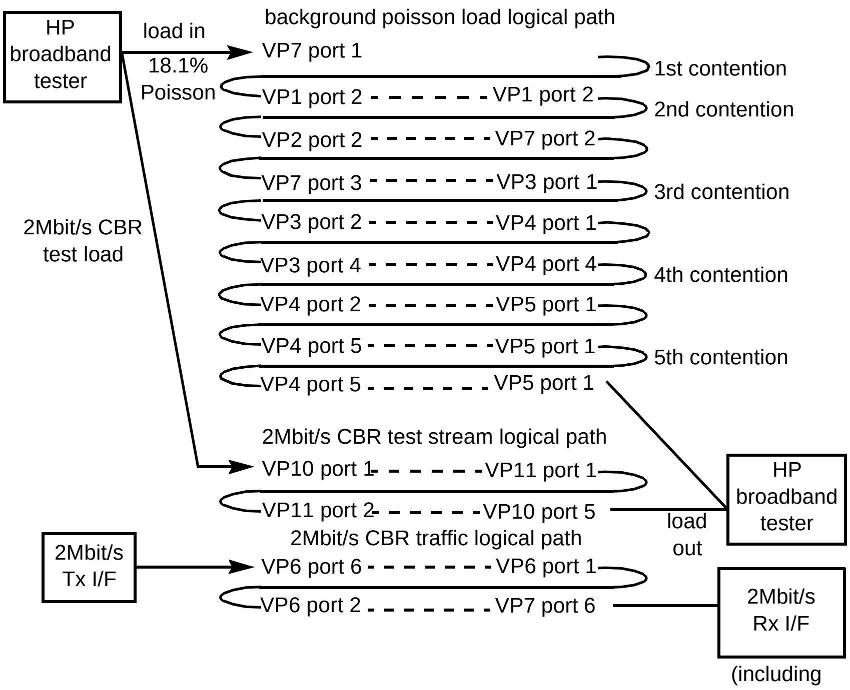


Fig 4 Logical assignment of VPs during testing.

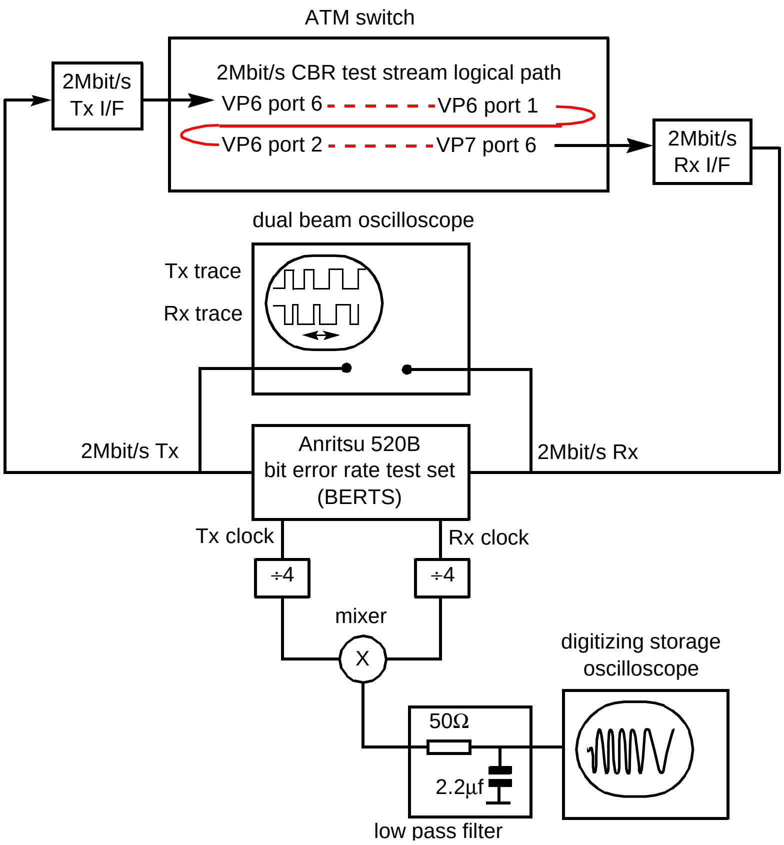


Fig 5 Test equipment configuration.

In order to calculate the magnitude and rate of phase change induced by a changing network load, the characteristic depicted on the lower digitising storage oscilloscope (shown in Fig 5) was measured. Here the reference transmit clock from the Anritsu BERTs set is mixed with the received clock. The mixer produces a single output cycle for each clock cycle traversed by the receive clock relative to the reference transmit clock. Thus, the induced phase wander in the 2-Mbit/s receive clock can be measured simply by counting the number of cycles output from the mixer and measured on the oscilloscope on load switch-on. Better still, by plotting the increasing periodicity of the mixer output waveform, the rate of change and thus time constant of the adaptive clock recovery mechanism can be estimated and compared with the predicted response in Fig 2.

4. Measurement results

4.1 Embedded 2-Mbit/s CBR stream from the broadband tester.

The delay characteristics of the embedded 2-Mbit/s CBR stream generated by the broadband tester were measured using the broadband tester with the Poisson background load turned both on and off. The results are shown in Table 1.

2-Mbit/s delay (μs)			
	minimum	mean	maximum
Poisson 18.1% background load off	141.1	143.85	150.1
Poisson 18.1% background load on	140.1	160.99	287.1
Difference in mean delay (μs)		17.14	

The difference in mean delay in clock cycles (unit intervals) at 2.048 MHz = $17.14*10^{-6}*2.048*106$ = 35 UI which compares reasonably with that given by the worst case M/D/1 queuing model of 39 UI.

4.2 Direct measurement results of the 2-Mbit/s receive clock

The dual beam oscilloscope traces of Fig 5 were found to indicate clock wander in the order of 1 UI peak to peak from the 2-Mbit/s receive clock even in the absence of contending network load. This meant that the background phase noise itself would produce peak-to-peak output from the mixer which would obscure the expected mixer output measurement traces during load switching. It was therefore decided to include clock dividers (divide-by-four) prior to the mixer to reduce this background phase noise to below 1 UI/4 = 0.25 UI peak to peak, thus ensuring a much lower ‘noise’ output from the mixer. Given the presence of the dividers, for every 4 cycles (UI) of phase wander from the received network clock a single ‘sinusoidal’ cycle should now be expected from the mixer output. The measurement trace resulting from switch-on of the background load is depicted in Fig 6, and the trace resulting from load switch-off is depicted in Fig 7.

The total number of cycles in the mixer output waveform on load switch-on was seven and on load switch-off was eight. It can therefore be estimated that the increase in network load causes an approximately 8*4 = 32 UI phase wander. This compares well with the delay measurement of the 2-Mbit/s background load given the uncertainty of up to 8 UI caused by the use of the clock dividers and the response of the adaptive clock to rms rather than mean delay.

The time constant of the adaptive clock can be estimated from Figs 6(b) and 7(b) as approximately 0.8 s, which compares well with the simulation result of Fig 2.

This result shows that, as expected, the output phase of the 2-Mbit/s receive clock directly tracks slow delay changes due to network load. In the case measured the phase was shown to change by ~ 32 UI (clock cycles) for a load increase from ~ 0 to 93.2%.

Wander limits

The limit for long-term phase wander for a 2-Mbit/s PDH CBR stream, as defined by ITU G.823, is 36.9 UI at $1.2*10^{-5}$ Hz (daily wander limit) down to 18 UI between 0.01 Hz and 1.667 Hz. Given any change in network load with periodicity greater than the time constant of the adaptive clock’s phase-locked loop, the receive clock phase will directly track the rms switch delay. Thus, for example, assuming in the worst case the network load changes by 93.2% over a period of, say, one minute, the receive clock phase would change by ~ 32 UI against a downstream equipment tolerance limit of only 18 UI. Worse still, not all of this tolerance limit could be apportioned to the ATM network. It seems likely therefore that, for a network comprising a number of ATM switches passing through busy hours and idle periods, excessive wander, beyond the ITU G.823 limits, will be exhibited by the exiting CBR cell stream.

Jitter limits

The jitter limit for a 2-Mbit/s PDH CBR stream, as defined by ITU G.823, is 1.5 UI peak to peak between 20 Hz and 100 kHz. The amount of jitter above 20 Hz is limited by the rate of change allowed by the adaptive clock’s phase-locked loop. The amount of jitter that could be imparted by

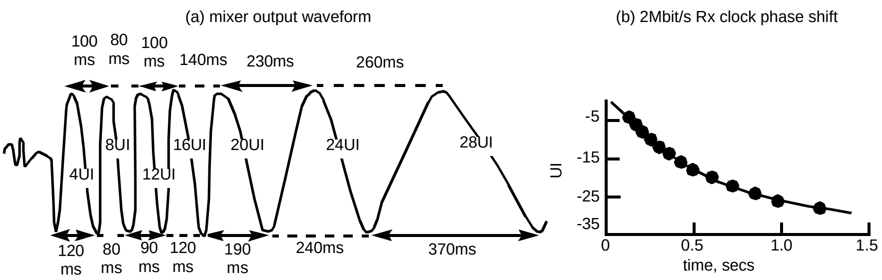


Fig 6 Poisson background load switch on-to-off.

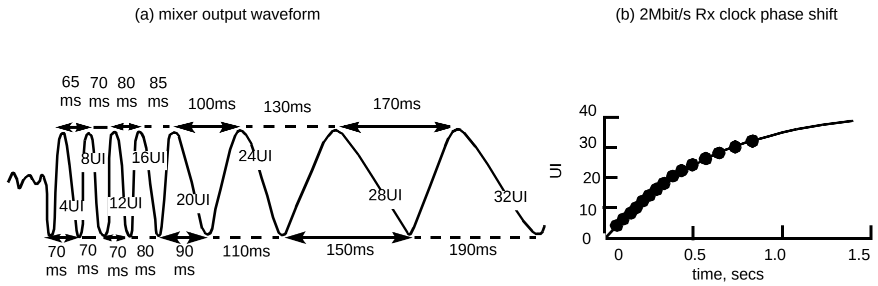


Fig 7 Poisson background load switch off-to-on.

a network load step change from ~ 0 to 93.2% can be estimated by passing the measured responses of Figs 6(b) and 7(b) through a 20-Hz high-pass filter. The results for Figs 6(b) 'Load switched on' and 7(b) 'Load switched off' are shown in Fig 8. Here the measured characteristic has been fitted to an exponential and the curve fit subjected to 20 Hz high-pass filtering. This result shows that even in the worst case a step change in network load will not cause excessive jitter (phase noise above 20 Hz) on the output stream.

5. Conclusions

The measurements performed on the ATM switch and 2-Mbit/s CBR transmission interface show that, for a single priority queue output port switch, the long-term phase of the 2-Mbit/s CBR output stream will follow the slow delay changes of the switch queue caused by varying network load. Given that the limits in ITU Recommendation G.823 for 2-Mbit/s wander lie between 36.9 UI and 18 UI, it

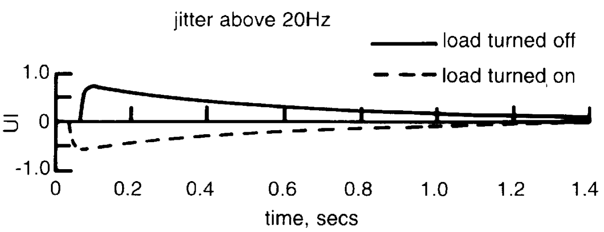


Fig 8 Increase in jitter with network load change.

appears impossible to guarantee such limits when using a single priority queue in association with adaptive clock recovery. This is particularly so both where the CBR streams are made to contend with large bursty data loads, and because not all of the downstream equipment wander tolerance limit can be apportioned to the ATM network. Thus any downstream equipment, which relies on the ITU G.823 limits for wander being met, could not be connected through such an ATM switch and adaptive clock recovery configura-tion and be guaranteed compliant wander behaviour.

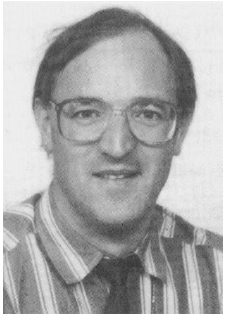
It may be possible to reduce excessive wander by use of a separate priority queue for CBR cell streams. For example, in the case considered where only one CBR stream exists, if this were to be serviced by a separate priority output buffer, it would suffer little CDV and thus cause virtually no wander at the output of the adaptive clock circuit. However, even here further study is required since the effects of waiting time jitter, the effects of dissimilar CBR rate contention at higher CBR loads, and contention with real time VBR video traffic, if placed in the same buffer, would need to be studied. In the absence of such studies either the synchronous residual time stamp (SRTS) method of timing delivery or network synchronous operation remains the preferred option for timing delivery of CBR services which require compliance with G.823 jitter and wander requirements.

The recommendation here is that, when single priority switch buffers are used in conjunction with adaptive clock recovery for delivering CBR streams, the CBR streams be used only for wander insensitive services.

References

1 Mulvey M, Kim I Y and Reid A B M: 'Timing issues of constant bit rate services over ATM', BT Technol J, 13, No 3, pp 35—45 (July 1995).

2 COST 224: 'Performance evaluation and design of multi-service networks', Ch 6.1.1, p 112 (1992).



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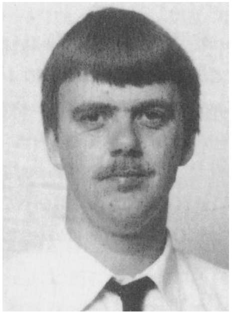

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